

Simulating 2368 temperate lakes reveals weak coherence in stratification phenology



Jordan S. Read^{a,*}, Luke A. Winslow^b, Gretchen J.A. Hansen^{b,c}, Jamon Van Den Hoek^d, Paul C. Hanson^b, Louise C. Bruce^e, Corey D. Markfort^f

^a Center for Integrated Data Analytics, U.S. Geological Survey, Middleton, WI, USA

^b Center for Limnology, University of Wisconsin - Madison, Madison, WI, USA

^c Wisconsin Department of Natural Resources, Bureau of Science Services, Madison, WI, USA

^d NASA Goddard Space Flight Center, Biospheric Sciences Branch, Code 618.0, Greenbelt, MD, USA

^e School of Earth & Environment, The University of Western Australia, Perth, Australia

^f IHR-Hydrosience and Engineering, Civil and Environmental Engineering, The University of Iowa, Iowa City, IA, USA

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ABSTRACT

Changes in water temperatures resulting from climate warming can alter the structure and function of aquatic ecosystems. Lake-specific physical characteristics may play a role in mediating individual lake responses to climate. Past mechanistic studies of lake–climate interactions have simulated generic lake classes at large spatial scales or performed detailed analyses of small numbers of real lakes. Understanding the diversity of lake responses to climate change across landscapes requires a hybrid approach that couples site-specific lake characteristics with broad-scale environmental drivers. This study provides a substantial advancement in lake ecosystem modeling by combining open-source tools with freely available continental-scale data to mechanistically model daily temperatures for 2368 Wisconsin lakes over three decades (1979–2011). The model accurately predicted observed surface layer temperatures (RMSE: 1.74 °C) and the presence/absence of stratification (81.1% agreement). Among-lake coherence was strong for surface temperatures and weak for the timing of stratification, suggesting individual lake characteristics mediate some – but not all – ecologically relevant lake responses to climate.

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1. Introduction

Lakes are diverse ecosystems that respond in complex ways to the stressors of climate change (Blenckner et al., 2007; Carpenter et al., 1992; Roach et al., 2013). While lakes have shown regional-scale coherence in temperature (see Livingstone, 2008), biological features and dynamics often are unrelated in neighboring lakes (e.g., Soranno et al., 1999), confounding our understanding of how lakes respond to climate drivers. Interpreting dissimilar responses of lakes to current and future climate is one of the grand challenges of modern limnology.

Lake-specific properties (such as morphometry and surrounding land cover) are recognized controls on the structure and function of aquatic ecosystems, but there is disagreement as to the role they play as mediators of climate variability. Despite large diversity in lake properties, analyses of lake surface temperatures

have established strong evidence for among-lake coherence to large-scale climate (Benson et al., 2000; Livingstone and Dokulil, 2001; Palmer et al., 2014). Conversely, coherence weakens or disappears below the lake surface (Kratz et al., 1998; Palmer et al., 2014), suggesting there is value in quantifying and explaining among-lake differences in responses to climate. At present, the diversity in lake responses to climate is not quantified; we lack the capacity to observe millions of lakes, and modeling approaches have not filled these information gaps.

Because temperature is a master factor for many aquatic ecosystem processes (Cardoso et al., 2014; Hanson et al., 2011; Magnuson et al., 1979; Paerl and Huisman, 2008), understanding thermal responses to climate is critical for predicting future biotic change (Sharma et al., 2007). Lake temperatures, stratification, and ice-cover durations are changing for many of the world's lakes (Livingstone, 2003; Magnuson et al., 2000; Schindler et al., 1990; Schneider and Hook, 2010), and these changes impact the timing and function of lake ecosystem processes (O'Reilly et al., 2003; Smol et al., 2005; Weyhenmeyer et al., 1999).

* Corresponding author. Tel.: +1 608 891 3922.
E-mail address: jread@usgs.gov (J.S. Read).

Recent advances in observational capabilities have allowed limnologists to address research questions at larger spatial scales (Heffernan et al., 2014; Soranno et al., 2010). Federal and state sensor networks, citizen science monitoring programs, and satellite/airborne observations have expanded considerably, contributing “Big Data” to aquatic ecology (Hampton et al., 2013; Keller et al., 2008). Satellite observations have documented lake warming (Schneider and Hook, 2010) and estimated water quality (Torbick et al., 2013). Gridded time-series data capture retrospective meteorological drivers (Mesinger et al., 2006; Mitchell et al., 2004) or predictions for future climate (Hostetler et al., 2011; Wilby et al., 2000) for large numbers of lakes. There are tremendous opportunities to leverage these environmental Big Data for a better understanding of the local context of lakes (e.g., Soranno et al., 2010), and broad-scale hydroclimatic interactions (e.g., Kucharik et al., 2000; Leonard and Duffy, 2013). Coupling mechanistic modeling with big data may provide means to assess among-lake diversity of lake responses to climate.

At present, two dominant process-based approaches exist for evaluating lake temperature responses at temporal scales appropriate for climate: simulations of generic lakes at broad spatial extents (Fang and Stefan, 2009; e.g., Hostetler and Small, 1999; Stefan et al., 2001), and detailed modeling of small numbers of real lakes (Cahill et al., 2005; Fang et al., 2012; Hadley et al., 2013). In order to quantify climatically driven regional dynamics and among-lake variability for large numbers of real lakes, a new approach that leverages the advantages of both simulation types is necessary.

Here, we examine the diversity of lake responses to climate by introducing a new methodology that uses freely available, large-scale datasets to run unique hydrodynamic simulations for thousands of lakes. Methods and tools were created explicitly to support future analyses of lakes and their design was informed by the research needs of the limnological community. In order to maximize community use and acceptance, all data, methods, and tools are freely accessible and openly shared. This study leveraged these elements to simulate an unprecedented 2368 lakes for 33 years and contrast individual and population-level lake responses to climate variability.

2. Materials and methods

2.1. Hydrodynamic model

Because our primary objective was to estimate temporal dynamics in water temperature profiles, a one-dimensional (1D) dynamical model was used to simulate each of the study lakes. The General Lake Model version 1.2.0 (GLM; Hipsey et al., 2013) was chosen for this study. GLM combines fluxes of mass and energy with a Lagrangian layer structure that adapts to changes in vertical gradients, and is freely available to all users. Many energy budget algorithms and mixing schemes in GLM are modern implementations of methods applied in other widely used 1D models (Hamilton and Schladow, 1997; Mooij et al., 2010).

Heat flux is the primary driver of lake temperature (Wetzel and Likens, 2000), and components of this flux were formatted as time-series input to GLM for the variables of wind speed, air temperature, relative humidity, precipitation, and downwelling long- and shortwave radiation (Fig. 1). At each time step, GLM accounts for energy fluxes into (e.g., downwelling radiation) and out of (e.g., evaporation) the lake, and propagates the resultant temperature changes to various layers according to heat transfer and vertical mixing algorithms. Detailed formulations of these numerical routines can be found in Hamilton and Schladow (1997) or Hipsey et al. (2013), which also includes a full parameter list for GLM with recommended default values that are based on field and laboratory

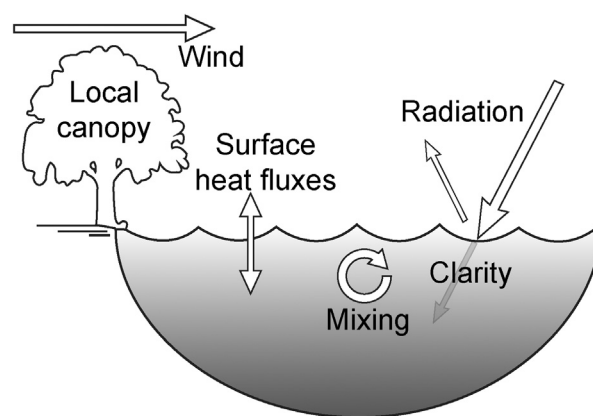


Fig. 1. A schematic of drivers and controls on lake temperature and stratification for a one-dimensional hydrodynamic model.

studies. Simulations did not include the influence of surface or groundwater flows, as coupling with hydrologic model output was beyond the scope of this study, but the impact of this omission on water temperature simulations was evaluated.

2.2. Model parameterization

Lake-specific data were used for static model inputs of hypsography, water clarity, and local wind sheltering (see below for details on how these were calculated or measured). The effects of these parameters on the hydrodynamic model are as follows: (1) hypsography influences size and depth dependent processes in the model, including the number of vertical layers used by the model, (2) water clarity controls the rate at which the visible wavelengths of solar radiation are attenuated in the water column, and (3) wind sheltering reduces the amount of energy from wind that is available for vertical mixing. Since a generic model parameterization that could be extended to other regions and lake types was desired, the remaining physical coefficients for GLM that parameterize equations of energy and momentum fluxes were set to default values.

The effect of water clarity on water temperature is parameterized in GLM according to the extinction coefficient for shortwave radiation (named ‘ K_w ’ in GLM, but we use the more common notation of ‘ K_d ’). The K_d parameter characterizes the exponential decay of light in the water column on a per meter (m^{-1}) basis. The parameter is used to define available light as $E(z) = E_0 \exp(-K_d z)$, where E_0 is the light energy at the surface and $E(z)$ is the remaining energy at depth z . As light is absorbed in the water column, it is converted into thermal energy and warms the waters. Thus, the general thermal effects of a smaller value of K_d (indicative of clearer waters) is to increase the amount of warming of deeper waters and decrease the warming of near-surface waters (Read and Rose, 2013).

Special attention was paid to the parameterization of mixing energy derived from wind speed because canopy and topographic features near the lake edge have a strong influence on physical dynamics in small and medium sized lakes (Markfort et al., 2010). The parameter in GLM that controls the flux of wind-driven mixing energy is the bulk aerodynamic momentum transfer coefficient, or C_M . The C_M parameter for each lake was scaled according to a cubic relationship with the wind sheltering coefficient (W_s ; Hondzo and Stefan, 1993; Markfort et al., 2010), calculated as $C_M = 0.0013 W_s^{1/3}$. Local canopy, topography, and lake size were included in the formulation of wind-sheltering coefficients for each lake, which were then converted to C_M values (see Van Den Hoek et al., *in review*). Contrary to many other multi-lake modeling efforts that specifically

tuned W_s to reduce model error (e.g., Fang et al., 2012), this coefficient was calculated as

$$W_s = \frac{2}{\pi} \cos^{-1} \left(25 \times h_s \sqrt{\frac{\pi}{A}} \right) - \frac{50 \times h_s}{A\sqrt{\pi}} \sqrt{A - 625 \times h_s^2 \times \pi}$$

where h_s is the average elevation difference between the top of the surrounding canopy and lake surface (m), and A is the surface area of the lake (m^2). This relationship was formulated according to equation 8 of Markfort et al. (2010). Lakes with a smaller wind sheltering coefficient have a reduced amount of vertical mixing energy available in the water column, and are often more strongly stratified.

2.3. Ice-cover simulations

Preliminary simulations for these lakes were continuous during the time period 1979–2011, but substantial bias in ice-cover duration using GLM's dynamical ice algorithms was discovered that also influenced the timing of springtime lake warming. To remove this bias, start and stop dates for every year were estimated for each lake using the methods of Shuter et al. (2013) for “Freeze-up model” and “Break-up model 1”, which were calibrated to ice-cover data from Wisconsin lakes (Benson and Magnuson, 2000). Initial temperature profiles were 4°C for each year.

2.4. Data sources

Publicly available datasets were used exclusively to parameterize and drive the individual lake models in order to support the reusability of these methods. Datasets with continental or global coverage were used whenever possible, including the North American Land Data Assimilation System (NLDAS-2; ldas.gsfc.nasa.gov/nldas), the National Elevation Dataset (NED; ned.usgs.gov), the ASTER GDEM2 30 m topographic elevation model (asterweb.jpl.nasa.gov), and Landsat 30 m imagery (landsat.usgs.gov) (Fig. 2). These products were used to generate lake-specific parameters or time-series for meteorological forcing, wind sheltering, and lake water clarity (Sup. Table 1).

The lakes chosen for this analysis had at least one fisheries survey or stocking recorded event by the Wisconsin Department of Natural Resources (WI-DNR). Spatial boundaries for the 2368 lakes used here were extracted from the Wisconsin hydrolayer, a statewide, 1:24,000 surface-water dataset (Sup. Table 1).

Depth and clarity data for the 2368 modeled lakes were available from various sources (Sup. Table 1; Table 1). Each study lake had data for maximum depth, and historical bathymetric maps exist for most lakes (dnr.wi.gov/lakes/maps). Hypsography (the relationship between depth and lake area) for 149 randomly chosen lakes was manually digitized from scans of these historical maps. For the remaining lakes, hypsography was estimated as a cone parameterized by the lake surface area and maximum depth. The shoreline development factor (SDF; Kent and Wong, 1982) was calculated for each lake as the ratio of the lake perimeter to the perimeter of a circle with equal area. The SDF is a metric of shoreline complexity, and a high SDF may be indicative of complex lake morphometry that would not be captured in a 1D lake model. Lake water clarity from WI-DNR and lakesat.org provided satellite-derived clarity estimates of Secchi depth (ZSD; see Torbick et al., 2013 for methodology). In situ measured Secchi depths from by WI-DNR monitoring programs and satellite Z_{SD} were converted to a light attenuation coefficient (K_d) using $K_d = 1.7/Z_{SD}$ (Poole and Atkins, 1929). K_d for the model simulations was used as the average of all observations for a given lake. When K_d was unavailable (145 of the 2368 lakes), the median was used (0.7 m^{-1} ; Table 1).

Table 1
Lake properties for all simulated lakes, lakes that were used for model validation, and a single lake (Big Muskellunge) used to compare with the population of lakes. K_d is the light attenuation coefficient and SDF is the shoreline development factor, a measure of shoreline complexity. Format shown for multiple numbers is median (25th quantile, 75th quantile) or [minimum, maximum].

Lake dataset	n (#)	Area (km^2)	z_{max} (m)	K_d (m^{-1})	h_s (m)	SDF (-)	Elevation (m)	Latitude ($^\circ\text{N}$)	Longitude ($^\circ\text{E}$)
Simulated lakes	2368	0.308 (0.119, 0.805)	6.71 (3.66, 10.7)	0.699 (0.498, 1.07)	7.24 (5.62, 9.13)	1.65 (1.33, 2.20)	[177.27, 558.01]	[42.497, 46.849]	[−92.770, −86.941]
Validation lakes	434	0.952 (0.435, 2.18)	9.14 (6.10, 14.0)	0.780 (0.503, 1.18)	7.64 (6.11, 9.39)	1.89 (1.52, 2.58)	[180.56, 536.15]	[42.507, 46.387]	[−92.744, −86.986]
Big Muskellunge	1	3.96	21.34	0.36	9.4	2.39	500.88	46.017	−89.615

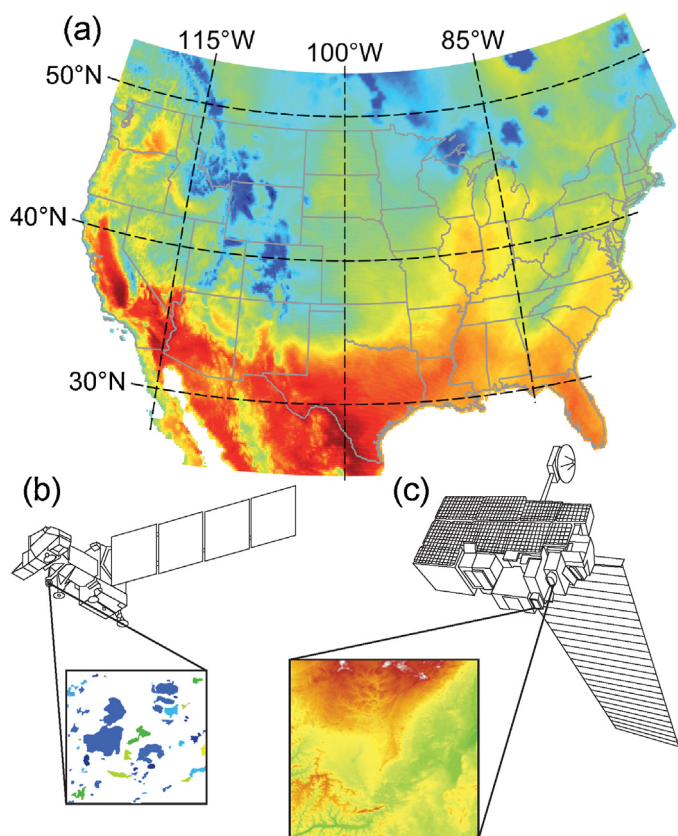


Fig. 2. Continental and global-scale data sources are critical inputs for parameterizing and driving thousands of one-dimensional lake models. (a) The NLDAS-2 hourly gridded data for air temperature on June 1st 1979. (b) Landsat imagery was used to estimate water clarity (clarity indices provided by lakesat.org). (c) Topographic elevation data from the ASTER GDEM2 digital elevation model were used to estimate wind sheltering heights within lake buffers.

In this study, 434 lakes had observational data that could be used to validate model performance. Data from citizen monitoring, WI-DNR, and the North Temperate Lakes Long-Term Ecological Research program (NTL-LTER; lter.limnology.wisc.edu) were combined into a water temperature validation dataset containing 140,148 unique measurements. These validation lakes were representative of the various lake types and geographical regions in the full population of simulated lakes (Table 1). A single lake, Big Muskellunge Lake (BM), was also examined in detail. BM is a core study lake in the NTL-LTER program, with data collection beginning in 1981. BM was not chosen to be representative of the remaining lakes, but instead to examine the applicability of extending patterns in climate responses from frequently monitored “sentinel lakes” (such as BM) to the more diverse population of lakes.

2.5. Web geo-processing

Gridded datasets for meteorological forcing and static parameters were spatially weighted and converted into lake-specific GLM inputs. Re-projecting, sub-setting, and transforming gridded data according to overlap with irregular geographical features (i.e., lakes) involves challenging algorithm development and data management techniques. This study leveraged web processing algorithms to reduce these burdens. Two representations of lakes and their surroundings were used to support two different regions of interest for data subsampling: the “lake” was the original polygon from the WI hydrolayer, and the “lake-buffer” was a 100 m wide buffer around the lake. Shapefiles representing all lakes and lake-buffers were hosted on sciencebase.gov, which has an

automated web feature service (WFS) extension for shapefiles. WFS is a standard for web-enabled exchange of spatial data that supports filtering of spatial data based on feature attributes.

Following the creation of WFSs for each of the two feature datasets, a geospatial web processing service was used to calculate lake-specific data from the gridded meteorology, canopy cover, and topographic data for each lake. The Geo Data Portal (GDP; [Blodgett, 2011](#)) is designed to automate the processing of gridded data according to feature geometries that may differ in spatial resolution and/or native spatial reference systems, and all processing and data exchange are completed via the Web. The GDP extracts data from input source data and returns a post-processed, ASCII file of the results filtered according to the lakes in each WFS. GDP-outputted results were then used either to parameterize the GLM model runs or as model driver data.

2.6. Model post-processing

Lake simulations were not calibrated to individual lakes, and simulation results were validated against observational data where available. Because observational data were typically available at a lower frequency than daily modeled temperatures, no phenological metrics were used for validation (e.g., the onset of thermal stratification). Four main metrics were used to evaluate simulation quality in terms of goodness of fit (root mean squared error or R^2): Schmidt Stability, epilimnion, hypolimnion, and the full water temperature profiles. Model results were processed and distilled into time-series files of water temperatures and physical metrics. The full water column temperature profiles included a direct comparison between all measured water temperatures (at all depths) and their depth-subsampled equivalents from model outputs. Epilimnion and hypolimnion temperatures were the volumetrically averaged temperatures for each of the two thermal layers, and were calculated only when the lake was stratified. Schmidt Stability was also calculated according to [Read et al. \(2011\)](#). Additionally, the model’s representation of stratification was evaluated based on the rate of false absence and false presence compared to observations. Lakes were considered stratified when the difference between surface and bottom water temperatures exceeded 1 °C ([Stefan et al., 1996](#); [Woolway et al., 2014](#)).

2.7. Sensitivity analysis for selected model parameters

Although a detailed uncertainty analysis for all lake modeling inputs was beyond the scope of this paper, the sensitivity of hypolimnetic temperatures to key model parameters was evaluated. Simulated hypolimnion temperatures often have the highest errors compared to shallower depths ([Weinberger and Vetter, 2012](#)), and we examined the sensitivity of modeled temperatures to water clarity, lake residence time, and wind sheltering height. All three of these parameters are known to have an influence on hypolimnetic water temperatures ([Killworth and Carmack, 1979](#); [Markfort et al., 2010](#); [Read and Rose, 2013](#)), and were either parameterized via remote sensing data or (in the case of residence time) neglected as model inputs. Using the summer (July–September) mean bottom water temperatures as a response variable, lake-specific parameters were varied within a range of $\pm 50\%$ of their original values.

In the case of residence time, which was not used in the original simulations, an estimate of residence time for each lake (WI-DNR; M. Diebel, *unpublished*) was used to estimate inflow and outflow volumes while holding lake volume at steady state. Daily inflow volume, inflow temperature, and outflow volume (outflow temperature is calculated by the model) were added to each lake model, and modified by $\pm 50\%$ for the sensitivity analysis. Because inflow temperatures of surface waters and groundwater were



Fig. 3. Lake modeling framework includes (1) selecting a lake feature (the surface shape of Big Muskellunge Lake; WI, USA is shown here in blue for example purposes) and gridded data, (2) extracting lake-specific data using the Geo Data Portal, (3) translating time-series and static results from the GDP into model input, (4) driving the lake simulation model, and (5) processing model results for each simulation. The Big Muskellunge Lake outline is used as an example. (For interpretation of the color information in this figure legend, the reader is referred to the web version of the article.)

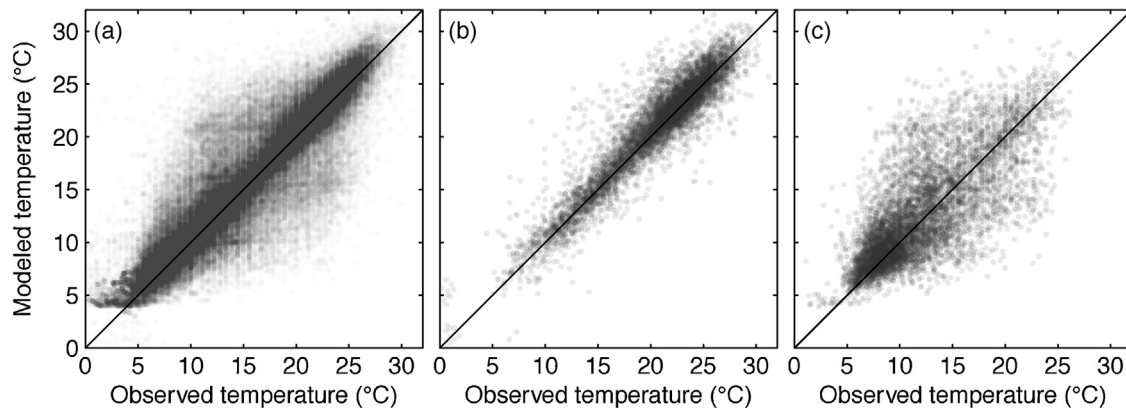


Fig. 4. Temperature validation for 434 lakes. (a) A comparison between model output and field temperature observations for all depths had a RMSE of 2.78 °C ($n = 140,148$). (b) Epilimnion comparisons had a RMSE of 1.74 °C ($n = 7240$), and (c) RMSE of 3.33 °C for hypolimnion temperatures ($n = 7240$).

unknown for the majority of these lakes, two scenarios were used to provide reasonable limits for these thermal inputs: (1) daily air temperatures were used as inflow temperatures, and (2) the seasonally-averaged air temperature was used as a constant inflow temperature.

2.8. Lake modeling framework

A framework was created for this study that automated the processes of data assimilation, data translation, model parameterization, model execution, and the calculation of derivative model results for each lake (Sup. Table 2; Fig. 3). R version 3.0.1 (R Development Core Team, 2013) was chosen as the scripting language to support these efforts because it is platform independent and open-source, making it particularly conducive to sharing. Documented R packages were developed to facilitate each of the following: automated geo-processing (geoknife), parameterization and analysis with the lake model (glmtools), and post-processing of model results and observational data (rLakeAnalyzer). Each individual R package includes detailed documentation and help files, but the purpose of each package in the modeling framework is described briefly below.

The geoknife package was created to simplify and automate the execution of GDP processing requests. The glmtools package simplifies interactions between users and the GLM executable model by providing methods for parameterizing the model and analyzing results data (including depth-based subsampling of output files). rLakeAnalyzer calculates derivative physical metrics from water column temperature profiles and standardizes comparisons between model and in situ observational data. rLakeAnalyzer is an R implementation of Read et al. (2011).

To speed up the iteration time of the modeling framework, modeling and processing were parallelized onto a HTCondor computer cluster (Thain et al., 2005). Using a pool of 26 compute nodes, this process reduced the modeling of all 2368 lakes for 33 years from multiple days (using a single computer) to a few hours.

Table 2

Pearson product-moment correlations between lake-specific root mean squared error and parameters.

Parameter	Correlation
$z_{\max}$	0.192 [*]
Area	0.027
h_s	0.029
K_d	−0.130 [*]
Elevation	−0.212
SDF	−0.039
Latitude	−0.247 [*]
Longitude	0.148 [*]

^{*} $p < 0.01$.

3. Results

3.1. Lake water temperature validation

The root-mean-square error (RMSE) between the model outputs and all temperature observations was 2.78 °C (Fig. 4a; $n = 140,148$), 1.74 °C for epilimnion temperatures (Fig. 4b; $n = 7240$), and 3.33 °C for hypolimnion temperatures (Fig. 4c; $n = 7240$). Model output and field observations agreed on the presence/absence of stratification 81.1% of the time ($n = 8690$). The rate of false positives (type I errors) was 12.9% ($n = 214$) and 20% for false negatives (type II errors; $n = 1811$). Twenty nine of the 434 validation lakes were responsible for over 50% of these mischaracterizations (type I and II errors). Comparisons for modeled versus observed Schmidt Stability had an R^2 of 0.93 ($n = 10,715$; data not shown).

The median RMSE on the full temperature data for individual lakes was 2.44 °C. Lake-specific RMSE was significantly correlated ($p < 0.01$) to several morphological and geographical parameters, including lake depth, elevation, latitude, and longitude, and water clarity (Table 2). Lake area, surrounding canopy height, and the shoreline development factor were not significantly correlated with lake errors (Table 2).

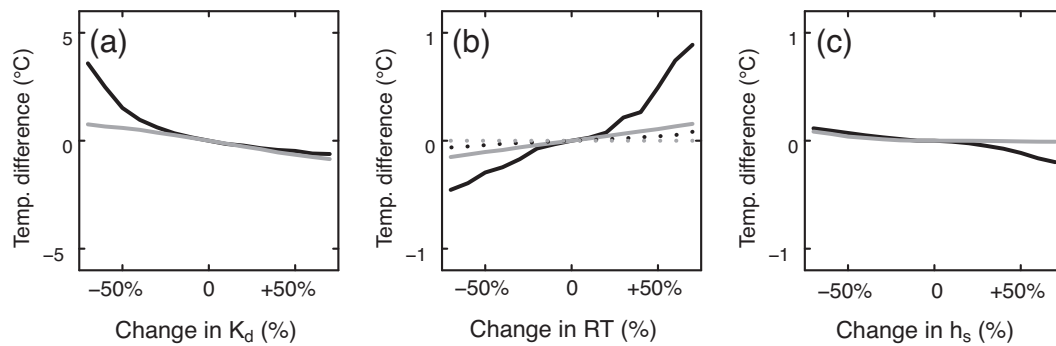


Fig. 5. Sensitivity of bottom water temperatures to changes in three model parameters for 1996 simulations. Median change in bottom water temperatures for lakes >100 ha (black line) and <100 ha (gray line), resulting from shifts in (a) the water clarity parameter (K_d), (b) residence time (dashed line is daily inflow temperatures equal to daily air temperatures, and solid line is constant inflow temperatures equal to yearly average air temperature), and (c) lake sheltering height.

3.2. Deep water temperature sensitivity to model parameters

Hypolimnetic temperatures had the highest errors among the metrics that were evaluated (see Fig. 4c), and the sensitivity of these deep water temperatures to model inputs for water clarity, lake residence time, and wind sheltering height were evaluated. Deeper water temperatures were more sensitive to changes in water clarity when compared to residence time and wind sheltering height (Fig. 5). Larger lakes were more sensitive to changes in all three parameters (Fig. 5; black line), but lake surface area was significantly positively correlated with maximum depth ($p < 0.01$), which influences the model's layer structure and mixing algorithms.

4. Discussion

4.1. Simulation quality

The model performed well across a diverse population of lakes during the 33 year simulation period (Fig. 4). Default values were used for most model parameters; with lake-specific values for the light attenuation coefficient, bulk momentum drag coefficient, hypsography, and meteorological drivers. Despite the use of an automated model setup that assimilated broad-scale data and various remotely sensed parameters, the 2.44 °C median RMSE for individual lakes is encouraging. This median lake RSME was only ~1 °C higher than a recent multi-lake modeling study where the investigators used local metrological observations and calibrated simulations individually for each of 28 lakes (median RMSE 1.47 °C; Fang et al., 2012).

Greater RMSEs for lakes with clearer waters were evaluated (see Section 3.2), highlighting the need for high quality data to quantify the K_d parameter. Errors in the value of this parameter from satellite-derived clarity measurements and the omission of important inter- and intra-annual variability in water clarity both likely contribute to substantial temperature errors in deeper waters. The relationship between lake errors with latitude and longitude (elevation was significantly positively correlated with latitude; $p < 0.01$) could be the result of a spatial bias in the input data, or simply an expression of the spatially heterogeneous distribution of Wisconsin lakes.

4.2. Model application: lake responses to climate variability

Water temperatures and the timing of stratification were highly variable among years over the 33-year simulation period. A comparison between model outputs from the warmest and coldest climate years based on seasonally averaged statewide air temperatures (Fig. 6a and b; 1998 and 1996 respectively; Wisconsin State

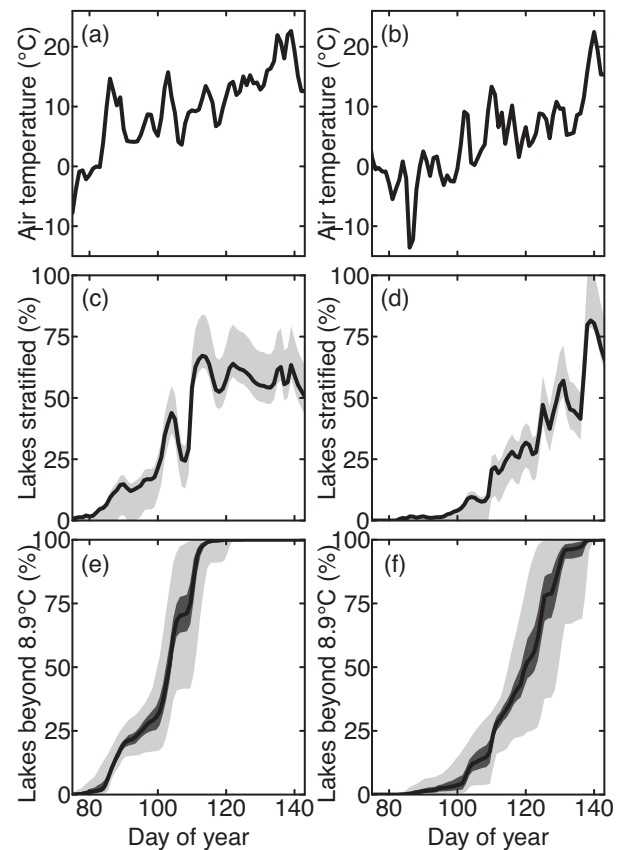


Fig. 6. Lake population dynamics for physical metrics during 2 contrasting years. Statewide averaged air temperatures for the warmest (a; 1998) and coldest (b; 1996) years during the simulation period (1979–2011). (c) and (d): Percentage of the 2368 simulated lakes that were stratified during a given day of year in 1998 and 1996, respectively. Extents for the gray bands for (c) and (d) represent the two extreme cases for stratification detection errors: only type II errors occurred, and only type I errors occurred. (e) and (f): percentage of lakes that exceeded the 8.9 °C walleye spawning threshold for a given day of year in 1998 and 1996. Gray and dark gray bands in (e) and (f) represent ranges in the threshold exceedance after randomly applying errors from Fig. 4b for the 95th and 50th percentiles, respectively.

climate office) revealed complexities in lake–climate interactions. For example, the date when at least half of the simulation lakes were stably stratified (a minimum of 7 days of sustained stratification) was 27 days earlier in 1998 than in 1996; but the maximum number of lakes stratified on any 1 day was greater for 1996 compared to 1998 (82.5% versus 71.4%; Fig. 6c and d).

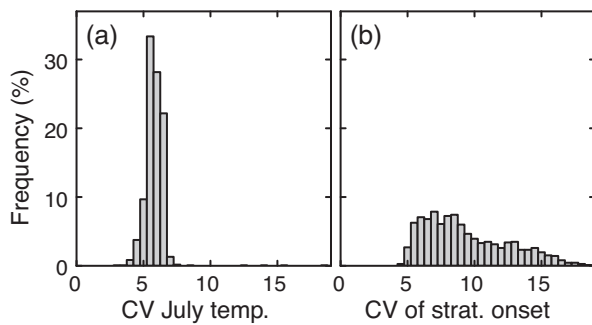


Fig. 7. Frequency distributions for lake-specific values of coefficient of variation (CV) for two physical metrics. (a) CV for 33 years of mean July surface water temperatures for each simulated lake ($n = 2368$). (b) Same as (a) but for the predicted day of year for the onset of stratification (only for lakes that stratified; $n = 1213$).

Multi-lake hydrodynamic model output can be used to predict ecosystem-level resilience to climate change, but applications with these data can also focus on responses at the species-level. An example is the examination of the different dates at which lakes exceeded 8.9°C during these 2 contrasting years. 8.9°C is an important temperature threshold for Wisconsin lakes, as it is cited as the temperature at which walleye, *Sander vitreus*, begin spawning (Wismer and Christie, 1987). The day of year when all 2368 simulated lakes were projected to pass this threshold was 19 days later in 1996 than in 1998 (Fig. 6e and f). Lake modeling outputs – similar to what we have highlighted here – can help fill observational gaps for physical conditions that may impact the health and function of lake ecosystems.

Individual lakes exhibited different degrees of variability in physical responses to regional climate, and the magnitude of these among-lake differences was dependent on the temperature metric used. Lake-specific variability was quantified by the inter-annual coefficient of variation (CV) for two climatically-relevant physical metrics during 1979–2011: summertime surface temperatures, which are an indicator of lake warming (Schneider and Hook, 2010; Sharma et al., 2007), and the timing of the onset of stratification, which is important for numerous ecosystem processes and hypothesized to be a sensitive indicator of climate change (Adrian et al., 2009; Winder and Schindler, 2004). The CVs for average July surface temperatures were similar for most lakes with little spread (Fig. 7a), suggesting that lake-specific properties have relatively little influence on how lakes respond to climate for this metric. Comparatively, the CVs of the timing of stratification differed greatly among lakes (Fig. 7b). These results suggest that climate variability may be differentially mediated for certain lake types, and that determining whether lakes in a region respond coherently or individually depends on the choice of physical metric.

4.3. Model application: among-lake coherence

These results may help investigators refine expectations for lakes in the future, but the modeling approach shown here also provides an ability to contrast the behavior of individual lakes with regional lake responses. An analysis of coherence was used to determine whether the variability shown in Fig. 7 is structured similarly among lakes or is an individualized expression of climate variability. We followed the methodology of Benson et al. (2000) for the calculation of among-lake coherence, which evaluated coherence and significance between lakes according to a pairwise Pearson product-moment correlation for the 33 years of simulation data. Big Muskellunge (BM) was chosen to provide a comparison with the other lake responses to climate because of the accuracy of BM model outputs (RMSE = 1.23°C) and the many long-term lake

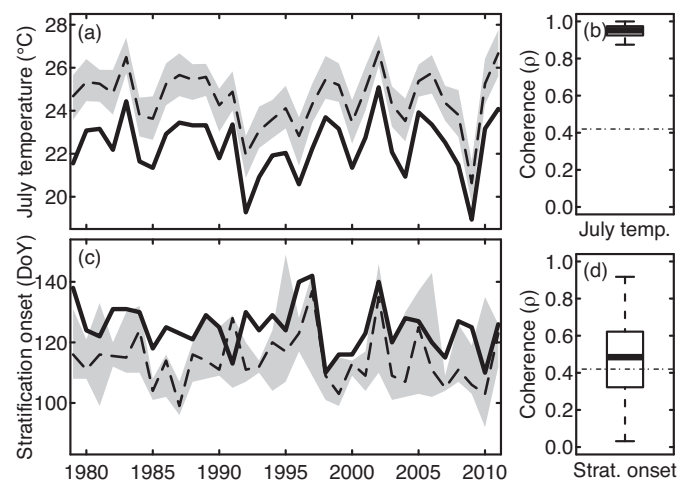


Fig. 8. Comparisons between temporal patterns for a single lake (Big Muskellunge Lake, WI USA; BM) and the lake population for two physical metrics derived from model output. (a) Average July surface water temperatures for BM (black line), the median of all lakes (dashed dark gray line; $n = 2368$), and the interquartile range (gray band). (b) Coherence between BM and all other lakes for July surface water temperatures. Dashed-dot line represents the critical value of ρ for significance in a one-tailed test ($p = 0.01$). (c) Day of year for the onset of stratification for BM (black line), the median onset date for all lakes that stratified (dashed dark gray line; $n = 1213$), and the interquartile range of stratified lakes. (d) Same as (b) but for the onset of stratification.

studies that include this frequently monitored lake (Benson et al., 2000; Kratz et al., 1998).

Big Muskellunge Lake surface temperatures were highly coherent with the rest of the simulated lakes, and its position among the population of lakes was relatively constant (Fig. 8a and b). In contrast, patterns in stratification were far less synchronous (Fig. 8c and d). These results support earlier work demonstrating greater agreement between temporal patterns for lake surface temperatures than for stratification metrics and the temperatures of deeper waters (Benson et al., 2000; Kratz et al., 1998; Palmer et al., 2014). Stratification patterns in Big Muskellunge and other frequently monitored lakes may not be representative of stratification dynamics for unobserved lakes that are exposed to the same regional climate.

Synchrony in lake surface temperatures and empirical studies that leverage these patterns may have provided limnologists with false confidence in our understanding of lake ecosystems in a changing climate. More ecologically relevant metrics (e.g., stratification and whole-lake temperatures) showed comparatively poor coherence (Fig. 8b versus d), and can express muddled or even divergent responses to climate variability (Benson et al., 2000; Tanentzap et al., 2008). Among-lake heterogeneity is particularly relevant for understanding how lakes respond to climate change and the types of lakes that will be most affected by climate change. Variability among lakes in response to climate change has important implications for species persistence and dispersal through the provision of microrefugia (Diffenbaugh et al., 2005; Keppel and Wardell-Johnson, 2012; Suggitt et al., 2011), and is deserving of additional study.

5. Conclusions

The General Lake Model (Hipsey et al., 2013) accurately predicted observed surface layer temperatures (RMSE: 1.74°C) and the presence/absence of stratification (81.1% agreement) when driven and parameterized by large-scale open data. Deeper water temperature simulations were less accurate (RMSE: 3.33°C for

hypolimnion temperatures), but there may be potential for future improvement. Our preliminary sensitivity analysis suggests that improving lake water clarity estimates may yield substantial gains in the quality of deep water simulated temperatures. Additionally, future improvements to the ice algorithms used in GLM are recommended to support continuous simulations for ice-covered lakes. Efforts were made to use the best possible data to parameterize and drive these lake models, but the quality of data varied among lakes and likely contributed to discrepancies in model accuracy. Despite these limitations, both the quality of models and publically available input data are expected to improve in the future.

Recent warming trends in surface temperatures are cause for alarm for many biota or ecosystem processes (Paerl and Huisman, 2008; Schneider and Hook, 2010; Sharma et al., 2007), but investigators should take caution in extending near-surface temperature patterns into predictions for climate responses of lake hypolimnion temperatures, stratification strength and timing, and whole lake temperatures. Our findings highlighted a contrast between strongly coherent surface temperatures and weakly coherent stratification patterns, suggesting that unified responses of lakes to regional climate may be unlikely for some ecologically relevant metrics. A mechanistic modeling approach, similar to what we have shown here (additionally, see Fang et al., 2012; Hadley et al., 2013; Weinberger and Vetter, 2012), can be used to refine our expectations for lake ecosystems in a changing climate.

Modeled lake temperature outputs generated at broad spatial scales for a large number of lakes could be useful for a wide range of practical applications. For example, hind-casted lake temperatures can be used to understand mechanisms behind observed changes in fish communities in these Wisconsin lakes, in addition to being used to explain variability in empirical data across the national landscape. Modeling individual lakes within the regional context allows investigators to capture spatial and temporal heterogeneity in individual lake responses. This variability is likely critical to understanding ecosystem responses to climate change that would be masked if “average” values that ignore spatial heterogeneity were used (Ackerly et al., 2010; Luoto and Heikkinen, 2008).

Similar to alternative 1D models, the utility of the hydrodynamic lake model used here can be extended by coupling additional modeling elements or input data sources. While beyond the scope of this study, combining these simulations with water quality models (Hamilton and Schladow, 1997) or terrestrial models that can dynamically model watershed processes (Kucharik et al., 2000) may improve our understanding of lake ecosystem function. Likewise, including measurements of stream flow (waterdata.usgs.gov) or hydrological modeling input/output could improve simulation quality for lakes with shorter residence times and potentially lead to improvements in stream temperature modeling efforts.

This modeling effort was developed with a foundation of internationally recognized standards for web processing and data transfer, providing a basic methodology that can easily be transferred to a range of different ecological research questions. Whether investigators could benefit from the exploration of thermal habitat change in the millions of lakes in the UGSG's national hydrography dataset (nhd.usgs.gov) or translate these methods into site-based terrestrial studies, the framework described here could be applied across a wide spectrum of modeling efforts. Additionally, statistically or dynamically downscaled climate projections could be easily exchanged for the NLDAS-2 forcing data and used to predict future scenarios for these ecosystems. These multi-system simulations represent a significant technical advance in limnology and ecological modeling. This framework will provide a useful tool in the interpretation of individual and population-level responses of ecosystems to climate.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ecolmodel.2014.07.029>.

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